WISER × Moderna Quantum Challenge 2026

RNA Secondary Structure Prediction via Quantum–Classical Hybrid Optimization

QUBO formulation • QAOA / VQE / Interference+Greedy solvers • Hierarchical divide-and-conquer scaling

Team QuantumCFO

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Executive Summary

Quantum-classical hybrid optimization reproduces classical minimum-free-energy (MFE) RNA structures with high fidelity on every tested instance. A Hadamard-transform interference decoder followed by classical greedy refinement ("Interference+Greedy") consistently outperforms QAOA and VQE on this problem class, matching the brute-force optimum on all 8 solver-comparison sequences and all 12 base validation sequences. A hierarchical divide-and-conquer solver bridges the gap between NISQ qubit budgets (~16 qubits) and the 313-qubit direct requirement of the 44-nucleotide challenge sequence, recovering its exact three-helix structure using at most 16 qubits per subproblem.

Headline results: 8/8 Interference+Greedy exact matches (comparison set) | 12/12 exact matches (base validation set) | 1.00 F1-score on the 44-nt challenge sequence | 40–75% qubit reduction via helix-level encoding

1. The Biological and Computational Problem

Biological context

Messenger RNA (mRNA) secondary structure influences molecular stability, translation efficiency, and manufacturability of mRNA-based therapeutics. Predicting the minimum free energy (MFE) fold — the thermodynamically most stable secondary structure — is a standard input to rational mRNA design. The number of possible folds grows exponentially with sequence length, making exhaustive classical enumeration intractable well before sequences reach practical length.

Computational formulation

RNA secondary structure prediction is formulated as a Quadratic Unconstrained Binary Optimization (QUBO) problem. QUBO coefficients are derived directly from ViennaRNA's thermodynamic model rather than hand-picked heuristic weights.

- Decision variables: one binary variable per chemically valid candidate base pair (i,j), subject to Watson–Crick pairing (A–U, G–C), wobble pairing (G–U), and a minimum hairpin loop size of 3 unpaired bases.
- Diagonal terms (loop-closure penalties): the real ViennaRNA energy of the structure with only pair (i,j) formed.
- Off-diagonal terms (stacking cooperativity): for adjacent nested pairs (i,j) and (i+1,j−1), a bonus equal to the real combined two-pair energy minus the two independent single-pair energies. This term is required, not optional — omitting it collapses the QUBO optimum to the fully unfolded structure regardless of sequence composition.
- Constraint penalties (cross-terms only): one-pair-per-base (λ=4.0) and no-pseudoknot (λ=6.0), each contributing a penalty only when both conflicting variables are actually selected together, not as an unconditional bonus. An earlier formulation applying constraint penalties unconditionally made thermodynamically unfavorable lone pairs appear artificially favorable; restricting penalties to cross-terms corrects this.

Quantum approaches evaluated

- QAOA — Quantum Approximate Optimization Algorithm, p = 2 layers
- VQE — Variational Quantum Eigensolver, hardware-efficient (RY + ring-CNOT) ansatz
- Interference+Greedy — Hadamard transform, phase encoding of the exact Ising cost Hamiltonian, classical greedy refinement

Pseudoknots are explicitly excluded from the model via the no-pseudoknot penalty term above; this scopes the formulation to nested (non-crossing) secondary structures, consistent with the standard MFE folding problem ViennaRNA itself solves.

2. Quantum Algorithms and Benchmarking

Benchmark: 10-nt prefix of the challenge sequence

The 10-nt prefix (GGAGCAAAAC) has an unfolded ViennaRNA MFE structure — all bases unpaired, E = 0.000 kcal/mol — against which each solver is benchmarked below:

Solver	Gap (kcal/mol)	F1	Runtime (s)
Brute Force	0.000	1.00	0.000
QAOA	0.000	1.00	1.40
VQE	3.000	0.00	0.53
Interference+Greedy	0.000	1.00	1.42

QAOA and Interference+Greedy both recover the exact MFE structure (F1 = 1.00, gap = 0.00). VQE does not, consistent with a barren-plateau-style trainability barrier in the RY+CNOT ansatz family on frustrated Ising problems. This was isolated separately: pure random parameter search performs comparably to COBYLA-optimized VQE, and matching the entangling topology to the cost Hamiltonian (all-to-all vs. ring) does not change the outcome.

Solver comparison: does quantum interference add value?

Hit-rate against the exact brute-force optimum, single stochastic trial per solver, across 8 sequences spanning 7–14 qubits:

Sequence	Qubits	Random Greedy	QAOA	VQE	Interference+Greedy
GCGCAUGCGC	8	1/5	1/1	0/1	1/1
GGGAUAUCCC	13	2/5	0/1	0/1	1/1
AUGCGCAUGC	7	5/5	1/1	0/1	1/1
GGCGCAAGCGCC	14	0/5	0/1	0/1	1/1
GCUGCAAAGCU	12	0/5	1/1	0/1	1/1
ACGGGCCACCG	10	1/5	1/1	0/1	1/1
GCGGCCACGCUA	14	0/5	0/1	0/1	1/1
GCCGUGAAGCGG	14	1/5	0/1	0/1	1/1

Interference+Greedy matches the brute-force optimum on all 8 sequences tested (8/8). QAOA matches on 4/8 in this run; VQE matches on 0/8. QAOA and VQE results come from a single stochastic trial each and vary run to run; Interference+Greedy has matched 8/8 in every run performed. Every Interference+Greedy-predicted structure also falls within ViennaRNA's suboptimal ensemble, an independent check of thermodynamic plausibility.

44-nt challenge sequence: hierarchical solve

Sequence: GGAGCAAAACUUGUCGAUUGAGAACAAAAUACAGAAUUUGCUUG (44 nt)

ViennaRNA MFE: .(((((((..((((...(((....)))...))))..))))))). (E = −7.90 kcal/mol)

Hierarchical solver result: .(((((((..((((...(((....)))...))))..))))))). (E = −7.90 kcal/mol) — exact match

Energy gap: 0.00 kcal/mol

F1-score: 1.00 (exact structural match)

Qubit requirement: 313 qubits under direct pair-level encoding; solved via divide-and-conquer with at most 16 qubits per subproblem

The recovered structure contains three nested helical regions enclosing a central interior loop, demonstrating exact recovery of a non-trivial multi-helix topology.

3. Scaling, Noise, and Resource Analysis

Quantum resource scaling

Qubit count vs. sequence length, pair-level encoding, real challenge sequence prefixes:

Length (nt)	Qubits	Exceeds 16-qubit budget
10	4	No
12	17	Yes
16	33	Yes
20	59	Yes
28	100	Yes
36	161	Yes
44	313	Yes

Qubit count scales roughly quadratically with sequence length under dense pairing. The 44-nt challenge sequence requires 313 qubits, exceeding current NISQ device capacity for exact statevector simulation.

Encoding comparison

Sequence	Length	Pair encoding	Partner-index encoding
AUGCGCAUGC	10	7 qubits	40 qubits
GCGCAUGCGC	10	8 qubits	40 qubits
GGGAUAUCCC	10	13 qubits	40 qubits
GGCGCAAGCGCC	12	14 qubits	48 qubits
44-nt example	44	313 qubits	264 qubits

Pair encoding exploits chemical sparsity and wins for short/medium sequences; partner-index encoding only wins asymptotically for long sequences, at the cost of harder multi-qubit equality constraints. A helix-level encoding — grouping contiguous stacked pairs into helical runs — reduces qubit count by 40–75% relative to the pair-level encoding (e.g. 14 → 6 qubits on a 14-nt sequence), and is the encoding used by the hierarchical solver below.

Noise robustness

Interference+Greedy hit-rate under depolarizing noise, 8-qubit instance (GCGCAUGCGC):

Noise level (p)	Hit-rate
0.00	1/1
0.05	1/1
0.10	1/1
0.20	1/1
0.30	1/1
0.50	1/1

Interference+Greedy maintains exact recovery at every tested noise level up to p = 0.50 on this instance. Density-matrix simulation (required for exact noise modeling) scales as O(4^m) memory versus O(2^m) for pure-state simulation, restricting exact noise studies to small instances (m ≤ 10 in the current environment).

4. Hierarchical Divide-and-Conquer Solver

To bridge NISQ qubit limits (~16 qubits) and the 313-qubit requirement of the 44-nt sequence, a recursive multi-anchor splitting algorithm is used:

- Candidate generation: identify maximal diagonal runs in a dot-plot matrix (helical candidates).
- Anchor selection: evaluate several candidate anchor helices, including sub-segments of maximal runs.
- Recursive decomposition: for each anchor, recursively solve the flanking (5′/3′) and interior regions.
- Energy-based selection: retain the anchor yielding the lowest total ViennaRNA energy for the reassembled structure.

Performance

- Base validation set (11 sequences): 12/12 exact matches (100%), including the 44-nt challenge sequence
- Stress test (5 constructed triple-nested-helix sequences): 1/5 exact matches (20%)

Limitation: bounded top-K anchor search is a heuristic, not exhaustive backtracking. A suboptimal anchor choice at one recursion level can foreclose the correct continuation at a deeper level, and such errors compound across levels. Closing this gap in general requires either exhaustive backtracking over anchor combinations or a dynamic-programming recursion over all split points (the Nussinov/Zuker approach) — neither of which is implemented here. Two secondary variants were also tested: substituting Interference+Greedy sampling for brute force at every leaf (use_quantum_leaf_solve=True) matches the brute-force baseline exactly on every sequence tested; ViennaRNA base-pairing-probability pruning (bpp_threshold) gives additional qubit reduction with no regression on the base set and raises the triple-helix stress-test score from 1/5 to 2/5.

5. Simulator Backend and Future Directions

Simulator backend comparison (14-qubit instance)

Backend	QAOA (60 iters)	Interference+Greedy (3000 shots)
default.qubit	4.49 s	0.039 s
lightning.qubit	1.55 s	0.010 s
Speedup	2.9×	3.9×

Tools and technologies

- Quantum SDK: PennyLane (lightning.qubit simulator)
- Classical RNA folding: ViennaRNA Python package
- Optimization: SciPy (COBYLA)

Future directions

- Hardware execution: validate the interference decoder on real quantum hardware.
- Error mitigation: zero-noise extrapolation and probabilistic error cancellation for noisy-device fidelity.
- Full Turner energy model: extend the QUBO formulation to internal loops, bulges, and terminal mismatches.
- Dynamic-programming integration: replace heuristic anchor search with an exact DP recursion over split points to guarantee optimality.
- Larger sequences: test the hierarchical solver on 100–200 nt sequences to assess scalability beyond the challenge example.

6. Limitations

- Anchor-selection heuristic in the hierarchical solver is not exhaustive and can compound errors on deeply nested multi-helix topologies (Section 4).
- VQE with a generic hardware-efficient ansatz shows a barren-plateau-style trainability barrier on this problem class and is outperformed by both QAOA and Interference+Greedy decoding.
- Exact noise-robustness studies are restricted to instances of ≤10 qubits by the $O(4^m)$ memory cost of density-matrix simulation.
- The current QUBO formulation excludes pseudoknots by design and does not yet model internal loops, bulges, or terminal mismatches beyond simple stacking.

7. Conclusion

Quantum-classical hybrid optimization reproduces classical MFE RNA structures with high fidelity on every tested instance. Interference+Greedy outperforms the variational approaches (QAOA, VQE) on this problem class, matching the brute-force optimum on all 8 solver-comparison sequences. The hierarchical divide-and-conquer solver bridges the scaling gap, recovering the exact structure of the 44-nt challenge sequence despite a 313-qubit direct requirement, while a constructed stress test (1/5 exact) identifies the specific topology class — deeply nested multi-helix structures — where the current anchor-search heuristic still falls short of a general guarantee. Interference-based quantum heuristics, combined with classical refinement and problem decomposition, are a practical direction for this problem class on near-term hardware.

8. Team Contributions

Team member	Role and contribution
Hisham Mansour	QUBO formulation and thermodynamic grounding (RNAQUBO/HelixQUBO); QAOA, VQE, and Interference+Greedy implementation; hierarchical divide-and-conquer solver.
Yasir Mansour	Scaling, noise-robustness, and solver-comparison studies; write-up and submission package.

9. Submission Package

- rna_fold_qubo.ipynb — full implementation and executed notebook (QUBO encodings, QAOA/VQE/Interference+Greedy solvers, hierarchical solver, benchmarking utilities), reproducing every result in this document
- This write-up (PDF/DOCX) and the accompanying slide deck
- Public GitHub repository with complete README (challenge, approach, methods, results, limitations, next steps, team contributions)

All sequences used are synthetic or drawn from the publicly provided challenge example; no proprietary Moderna data, patient data, or personally identifiable information is used at any point.